

# Lab1a - Bonus A

## Private EC2 + SSM Session Manager + VPC Endpoints (No SSH)

This runbook provides AWS Console step-by-step instructions to deploy and verify Bonus A. The goal is a private EC2 instance with no public IP, no SSH access, and private connectivity to AWS control-plane services using VPC endpoints.

### Design goals (what you are proving)

- EC2 is private (no public IP) and cannot be reached from the internet
- No SSH: access is via Systems Manager Session Manager
- Private subnets can reach AWS APIs via VPC endpoints (no NAT required for AWS services)
- Interface endpoints: ssm, ec2messages, ssmmessages, logs, secretsmanager (+ kms optional)
- Gateway endpoint: S3 (common gotcha in private environments)
- Least privilege IAM: GetSecretValue limited to your secret; GetParameter(s) limited to your path

### Assumptions

- Region: us-east-1 (use your lab region).
- You have permissions to create: VPC, subnets, route tables, endpoints, IAM roles, EC2, SSM, Secrets Manager, and Parameter Store entries.
- You will use Session Manager (no bastion, no SSH keys).

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# Architecture Overview

Bonus A places an EC2 instance in a private subnet with no public IP. The instance is managed via SSM Session Manager. VPC endpoints provide private access to AWS services without requiring NAT for AWS control-plane traffic.

| Component                         | Purpose                   | Notes                                                 |
|-----------------------------------|---------------------------|-------------------------------------------------------|
| Private EC2 instance              | Compute in private subnet | No public IP; inbound SSH closed                      |
| IAM role for EC2                  | SSM and limited reads     | AmazonSSMManagedInstanceCore + least-privilege policy |
| Interface endpoints               | Private AWS API access    | ssm, ec2messages, ssmmessages, logs, secretsmanager   |
| S3 gateway endpoint               | Private S3 access         | Attach to private route table                         |
| CloudWatch Logs                   | Audit + telemetry         | Logs can ship via logs endpoint                       |
| Secrets Manager + Parameter Store | Configuration             | Prove EC2 can read both via endpoints                 |

# AWS Console Steps

Follow these steps in order. Each step produces an artifact you can verify later.

## Step 1 - Create or verify the VPC

- 1 **VPC Console:** VPC → Your VPCs → Create VPC (or select existing).
- 2 Enable **DNS hostnames** and **DNS resolution** on the VPC.
- 3 Record the **VPC ID** for endpoints and verification.

## Step 2 - Create subnets (public and private)

- 1 **VPC Console:** VPC → Subnets → Create subnet(s).
- 2 Create at least one **public subnet** and one **private subnet** in the same VPC.
- 3 Launch Bonus A EC2 into the **private subnet**.

## Step 3 - Route tables

- 1 **Public route table:** add route 0.0.0.0/0 → Internet Gateway and associate with the public subnet.
- 2 **Private route table:** associate with the private subnet. Do not add an Internet Gateway route.
- 3 NAT is optional; endpoints cover AWS APIs. (Package installs may require NAT or a prebuilt AMI.)

## Step 4 - Create VPC endpoints (core requirement)

- 1 **VPC Console:** VPC → Endpoints → Create endpoint.
- 2 **Interface endpoints:** create one each for: ssm, ec2messages, ssmmessages, logs, secretsmanager (kms optional).
- 3 For each interface endpoint: choose **private subnets**, enable **Private DNS**, and attach an endpoint SG allowing inbound TCP 443 from the VPC CIDR.
- 4 **S3 gateway endpoint:** create and attach to the **private route table**.

## Step 5 - Create IAM role for EC2 (SSM + least privilege)

- 1 **IAM Console:** IAM → Roles → Create role → AWS service → EC2.
- 2 Attach managed policy: **AmazonSSMManagedInstanceCore**.
- 3 Add an inline policy restricting access to only:
- 4 • secretsmanager:GetSecretValue for your secret ARN
- 5 • ssm:GetParameter(s) for your parameter path (example: /lab/db/\*)

## Step 6 - Create configuration stores (Secrets + Parameters)

- 1 **Secrets Manager:** store a new secret (example: DB credentials). Record the secret name/ARN.

- 2 **SSM Parameter Store:** create /lab/db/endpoint (String) with a placeholder endpoint value.

## Step 7 - Launch the private EC2 instance (no public IP, no SSH)

- 1 **EC2 Console:** EC2 → Instances → Launch instance.
- 2 Network settings: choose your VPC and the **private subnet**.
- 3 Auto-assign public IP: **Disable**.
- 4 Security group: do not open port 22 (SSH). Keep inbound minimal.
- 5 IAM instance profile: select the role created in Step 5.
- 6 Launch and record the **Instance ID**.

## Step 8 - Connect using Session Manager

- 1 **Systems Manager:** Systems Manager → Session Manager → Start session.
- 2 If the instance appears, Session Manager is working via endpoints.
- 3 Start a session; use it for verification commands below.

## Student Verification (CLI Evidence)

Run commands from your local terminal unless noted otherwise.

### 1) Prove EC2 is private (no public IP)

```
aws ec2 describe-instances \  
--instance-ids <INSTANCE_ID> \  
--query "Reservations[].Instances[].PublicIpAddress"
```

Expected: **null**

### 2) Prove VPC endpoints exist

```
aws ec2 describe-vpc-endpoints \  
--filters "Name=vpc-id,Values=<VPC_ID>" \  
--query "VpcEndpoints[].ServiceName"
```

Expected includes: ssm, ec2messages, ssmmessages, logs, secretsmanager, s3 (kms optional).

### 3) Prove Session Manager path works (no SSH)

```
aws ssm describe-instance-information \  
--query "InstanceInformationList[].InstanceId"
```

Expected: your private EC2 instance ID appears.

### 4) Prove the instance can read both config stores (run inside SSM session)

```
aws ssm get-parameter --name /lab/db/endpoint  
aws secretsmanager get-secret-value --secret-id <YOUR_SECRET_NAME_OR_ARN>
```

### 5) Prove CloudWatch Logs delivery path is available via endpoint

```
aws logs describe-log-streams \  
--log-group-name /aws/ec2/<prefix>-rds-app
```

## Troubleshooting (Fast Fixes)

- **Instance not showing in Session Manager:** confirm EC2 has AmazonSSMManagedInstanceCore, endpoints exist with Private DNS enabled, and endpoint SG allows inbound 443 from the VPC CIDR.
- **Cannot reach Secrets/Parameters from SSM session:** verify IAM resource scope (secret ARN, parameter path) and confirm secretsmanager/ssm endpoints exist.
- **Package installs fail in private subnet:** without NAT, yum/apt may not reach public repos. Use a golden AMI, add NAT temporarily, or use internal mirrors. S3 gateway endpoint helps for S3-backed artifacts.
- **CloudWatch logs missing:** confirm logs endpoint exists and the instance is configured to ship logs (agent config).

## Submission Checklist

- Screenshot: EC2 instance details showing **No public IPv4 address**.
- Screenshot: VPC Endpoints list showing required services and the S3 gateway endpoint.
- Screenshot: Session Manager connected session.
- CLI output: describe-instances public IP query returns null.
- CLI output: describe-vpc-endpoints includes required services.